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COLORS AND BACKGROUNDS

OVERVIEW

- **CSS color names**
- **RGB and HSL color values**
- **Foreground and background colors**
- **Tiling background images**
- **More selectors and external style sheets**

Named Color Values

Specify foreground or background color using one of 140 predefined CSS3 **color names**:

```
h1 { color: red; }
```

```
h2 { color: darkviolet; }
```

```
body { background-color: papayawhip; }
```

learningwebdesign.com/colornames.html

Numeric Color Values

For more control, define colors numerically using one of these color models:

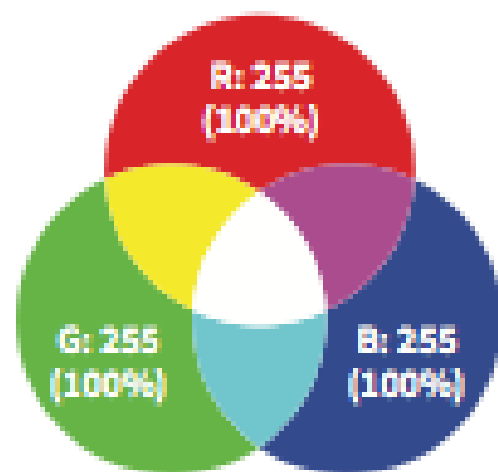
- **RGB** (combination of red, green, and blue light values)
- **RGBA** (RGB plus alpha transparency)
- **HSL** (hue, saturation, and luminosity)
- **HSLa** (HSL plus alpha transparency)

RGB Color

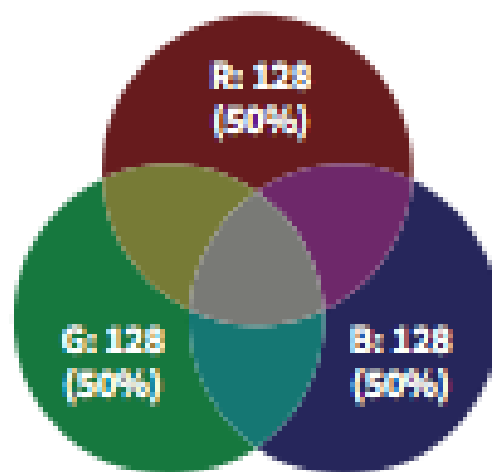
The **RGB color model** mixes color with red, green, and blue light.

Each channel can have 256 shades, for millions of color options.

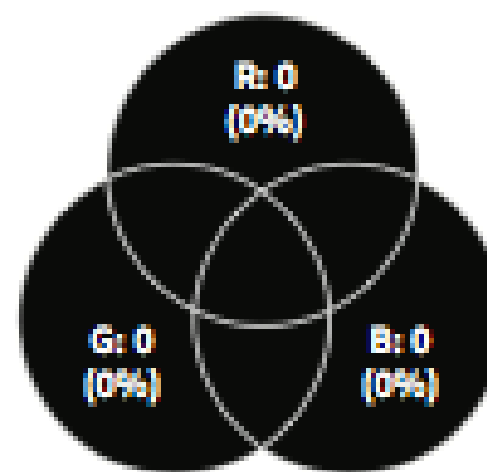
The RGB Color Model



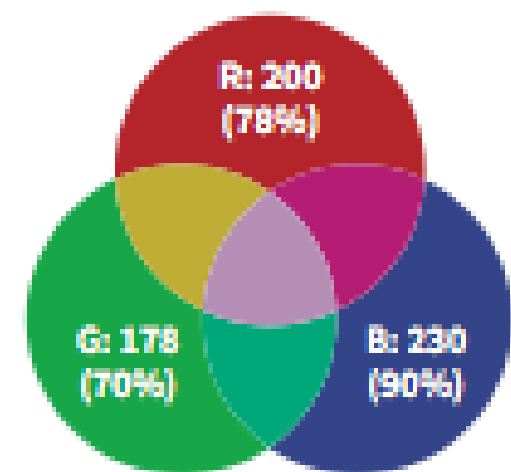
RGB: 255, 255, 255
white



RGB: 128, 128, 128
gray



RGB: 0, 0, 0
black



RGB: 200, 178, 230
pleasant lavender

RGB Values in Style Rules

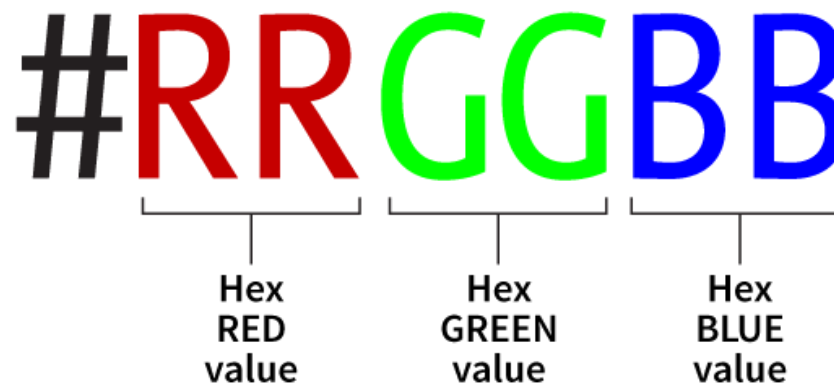
There are four formats for providing RGB color values:

- RGB values (0 to 255): **rgb (200 , 178 , 230)**
- Percentage values: **rgb (78% , 70% , 90%)**
- Hexadecimal values: **#C8B2E6**
- Condensed hexadecimal values (for double-digits only): **#F06** is the same as **#FF0066**

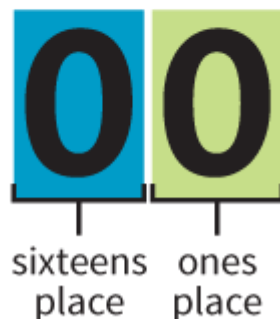
Hexadecimal RGB Values

Red, green, and blue values converted to **hexadecimal** and preceded by the **#** symbol.

Hexadecimal RGB values must be preceded by the # symbol.



Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F



The decimal number **32** is represented as

20

2 sixteens and 0 ones

The decimal number **42** is represented as

2A

2 sixteens and 10 ones

RGBa Color

- **RGB + an alpha channel** for transparency
- The first three values are RGB. The fourth is the transparency level from 0 (transparent) to 1 (opaque).

Playing with RGBa

Playing with RGBa

Playing with RGBa

```
color: rgba(0, 0, 0, .1);
```

```
color: rgba(0, 0, 0, .5);
```

```
color: rgba(0, 0, 0, 1);
```

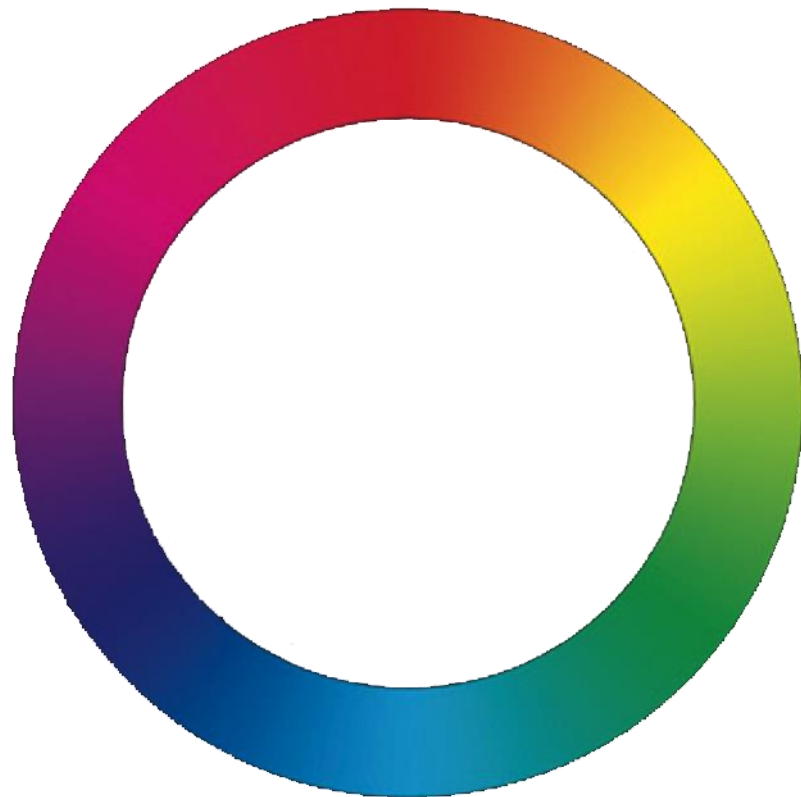

HSL and HSLa

- Colors described by values for **hue** (°), **saturation** (%), and **luminosity** (%):

hsl (180 , 50% , 75%)

- Hue specifies the position on a color wheel (in degrees) that has red at 0°, green at 120°, and blue at 240°.
- HSL is less commonly used than RGB, but some find it more intuitive.
- **HSLa** adds an alpha value for transparency.

HUE



```
1  hsl(0, 100%,50%)    /* red */
2  hsl(30, 100%,50%)
3  hsl(60, 100%,50%)
4  hsl(90, 100%,50%)
5  hsl(120,100%,50%)   /* green */
6  hsl(150,100%,50%)
7  hsl(180,100%,50%)
8  hsl(210,100%,50%)
9  hsl(240,100%,50%)   /* blue */
10 hsl(270,100%,50%)
11 hsl(300,100%,50%)
12 hsl(330,100%,50%)
13 hsl(360,100%,50%)   /* red */
14
15 hsl(120,100%,25%)   /* dark green */
16 hsl(120,100%,50%)   /* green */
17 hsl(120,100%,75%)   /* light green */
18
19 hsl(120,100%,50%)   /* green */
20 hsl(120, 67%,50%)
21 hsl(120, 33%,50%)
22 hsl(120,  0%,50%)
23
24 hsl(120, 60%,70%)   /* pastel green */
```

Foreground Color

color

Values: *Color value* (named or numeric)

Example: `blockquote {border: 4px dashed; color: green; }`

The **foreground** of an element consists of its text and border (if one is specified).

In the latitude of central New England, cabbages are not secure from injury from frost with less than a foot of earth thrown over the heads. In mild winters a covering of half that depth will be sufficient; but as we have no prophets to foretell our mild winters, a foot of earth is safer than six inches.

Opacity

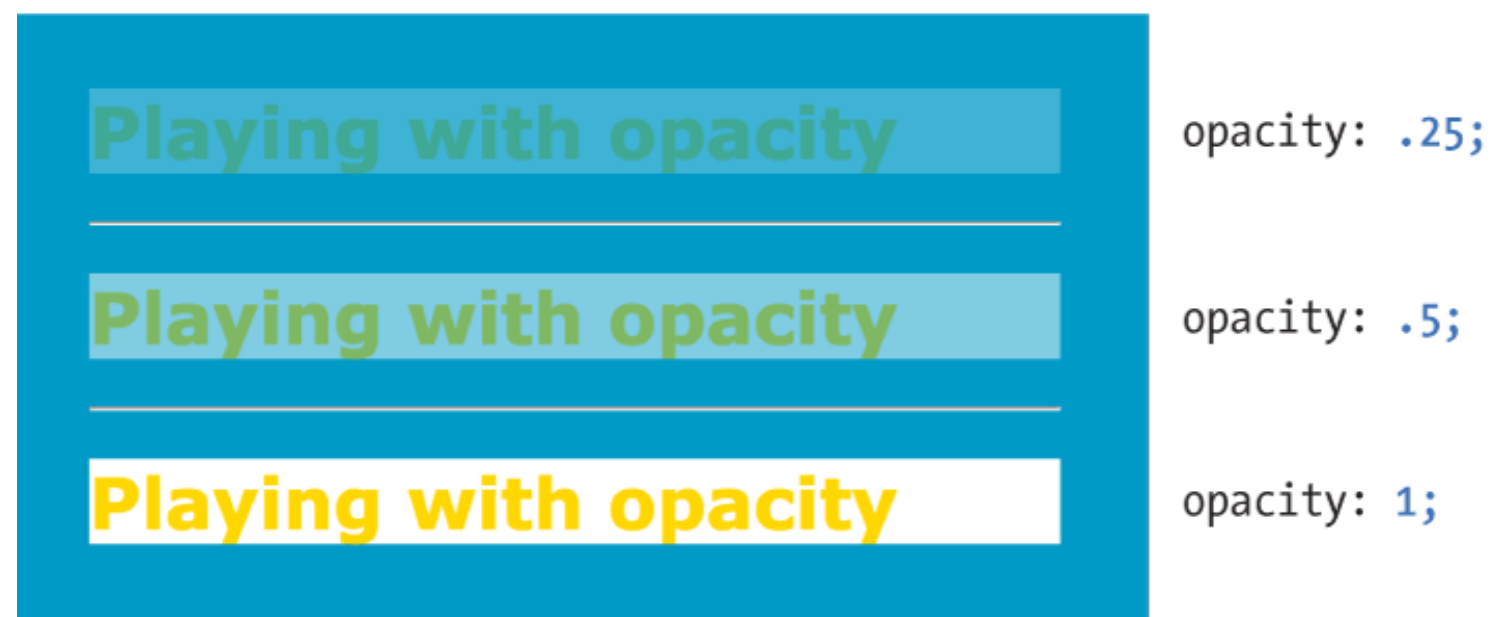
`opacity`

Values: *number* (0 to 1)

Example:

```
h1 {color: gold; background: white; opacity: .25;}
```

Specifies the transparency level from 0 (transparent) to 1 (opaque):



Tiling Background Images

background-image

Values: `url (location of image)` , `none`

Example: `body {background-image: url(star.png);}`

Locates an image to be used as a tiling background image behind an element. By default, it starts at the top, left corner and repeats horizontally and vertically:



star.png
150 × 150 pixels



Background Repeating

`background-repeat`

Values:

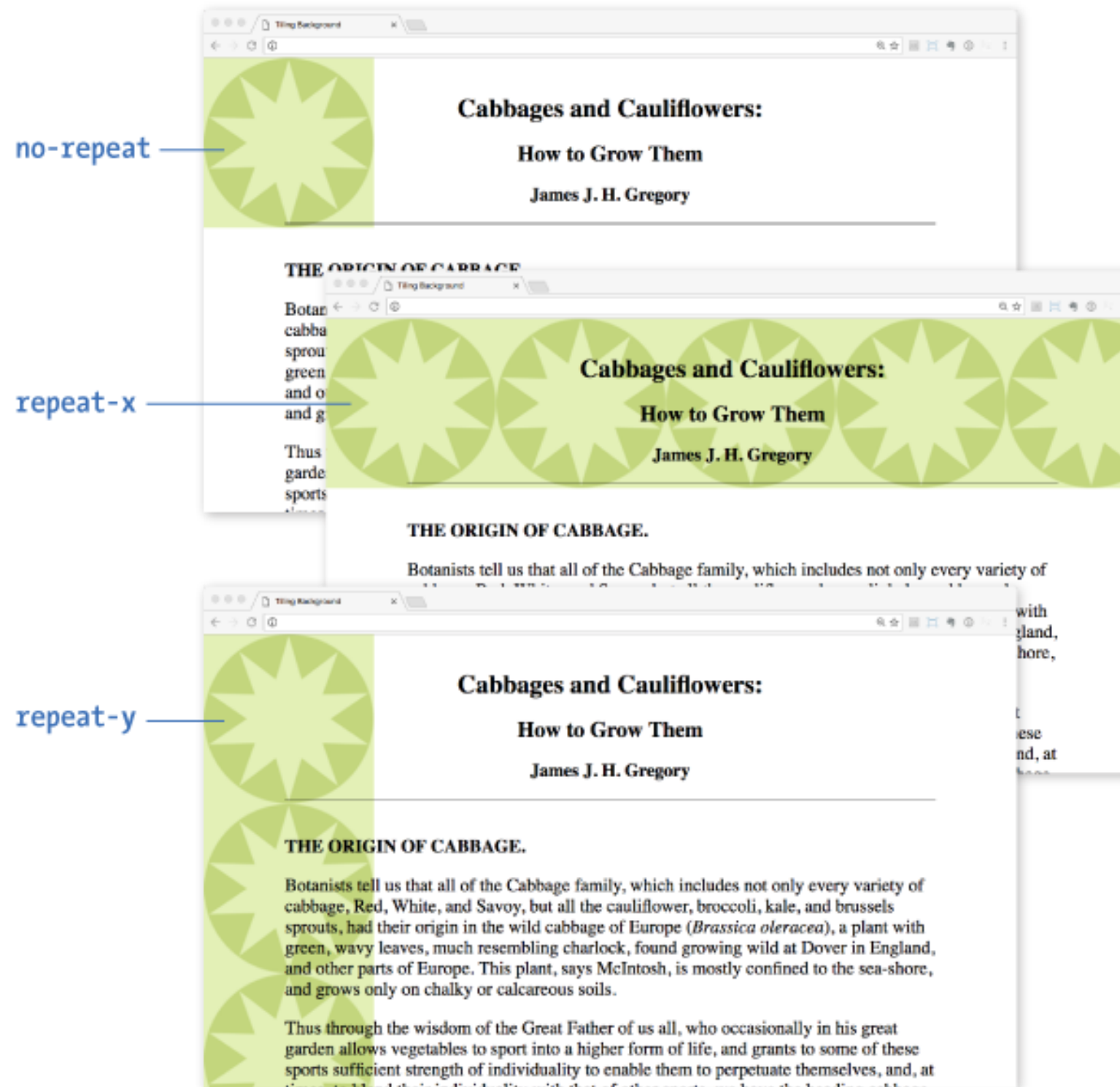
`repeat`, `no-repeat`, `repeat-x`, `repeat-y`, `space`, `round`

Specifies how the background image repeats and can restrict it to tiling in one direction or not at all:

- **`repeat-x`**: Tiles horizontally only
- **`repeat-y`**: Tiles vertically only
- **`space`**: Adds space around images so they fit in the window with no clipping
- **`round`**: Distorts the image so it fits without clipping

Demo: <https://developer.mozilla.org/en-US/docs/Web/CSS/background-repeat>

Background Repeating (cont'd)



Background Position

`background-position`

Values:

Length, percentage, left, center, right, top, bottom

Specifies the position of the **origin image**, the first image that is placed in the background from which tiling images extend.

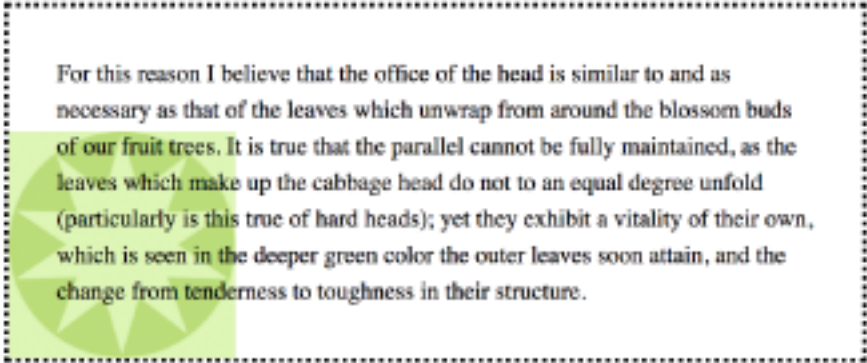
Examples (horizontal position goes first):

```
background-position: left bottom;
```

```
background-position: 300px 100px;
```

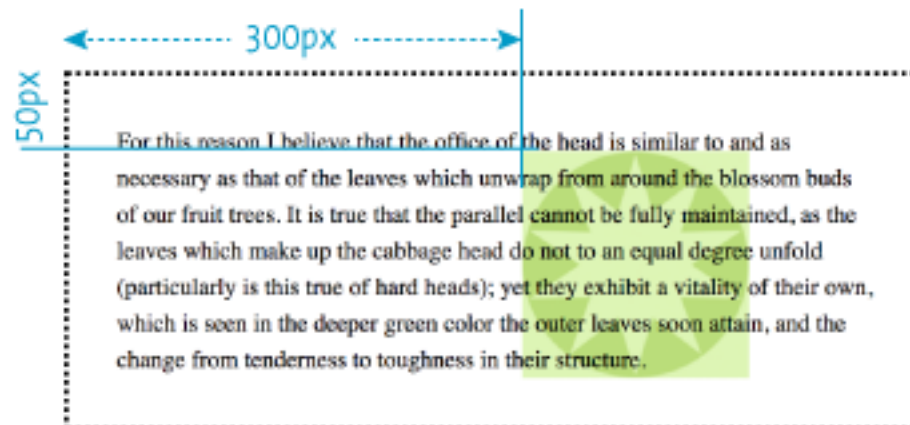
```
background-position: 25% 100%;
```


Background Position (cont'd)



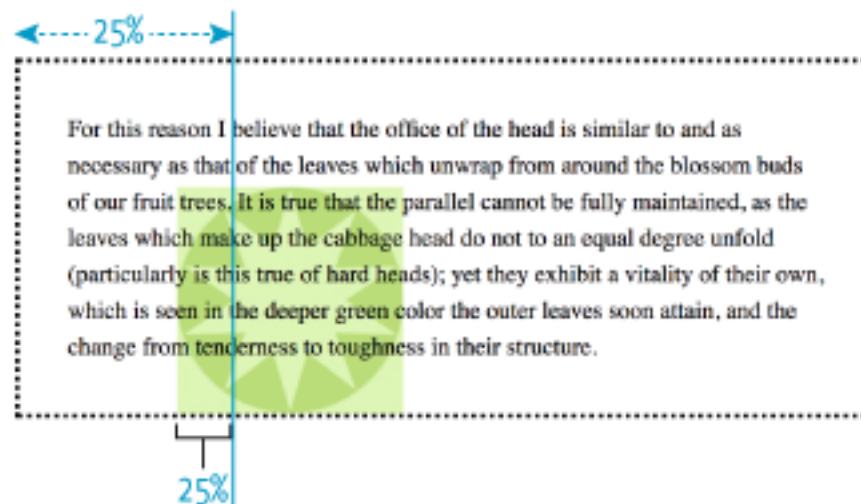
For this reason I believe that the office of the head is similar to and as necessary as that of the leaves which unwrap from around the blossom buds of our fruit trees. It is true that the parallel cannot be fully maintained, as the leaves which make up the cabbage head do not to an equal degree unfold (particularly is this true of hard heads); yet they exhibit a vitality of their own, which is seen in the deeper green color the outer leaves soon attain, and the change from tenderness to toughness in their structure.

`background-position: left bottom;`



For this reason I believe that the office of the head is similar to and as necessary as that of the leaves which unwrap from around the blossom buds of our fruit trees. It is true that the parallel cannot be fully maintained, as the leaves which make up the cabbage head do not to an equal degree unfold (particularly is this true of hard heads); yet they exhibit a vitality of their own, which is seen in the deeper green color the outer leaves soon attain, and the change from tenderness to toughness in their structure.

`background-position: 300px 50px;`



For this reason I believe that the office of the head is similar to and as necessary as that of the leaves which unwrap from around the blossom buds of our fruit trees. It is true that the parallel cannot be fully maintained, as the leaves which make up the cabbage head do not to an equal degree unfold (particularly is this true of hard heads); yet they exhibit a vitality of their own, which is seen in the deeper green color the outer leaves soon attain, and the change from tenderness to toughness in their structure.

`background-position: 25% 100%;`

Background Attachment

background-attachment

Values: `scroll`, `fixed`, `local`

Specifies whether the background image scrolls with the content or stays in a fixed position relative to the viewport.



A large non-repeating background image in the body of the document.



background-attachment: `scroll`;

By default, the background image is attached to the body element and scrolls off the page when the page content scrolls.



background-attachment: `fixed`;

When background-attachment is set to `fixed`, the image stays in its position relative to the browser viewing area and does not scroll with the content.

Background Size

background-size

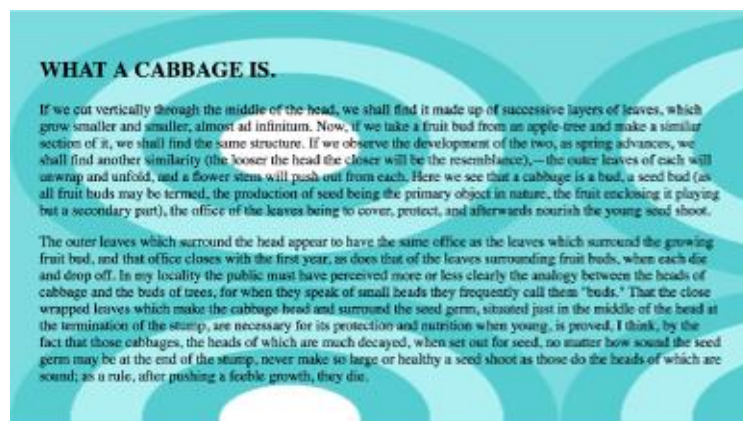
Values:

Length, percentage, auto, cover, contain

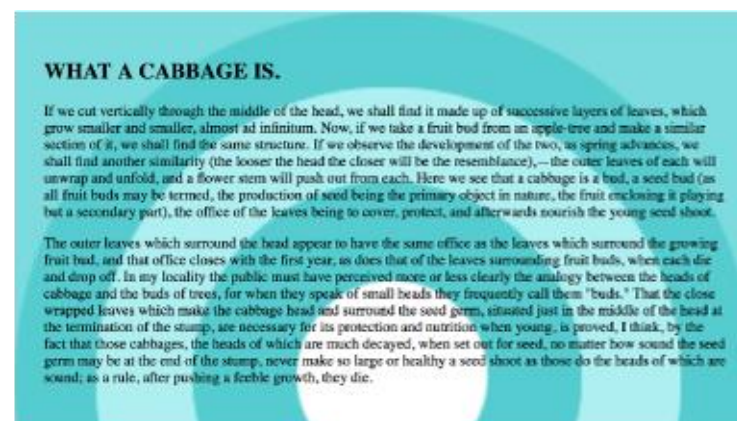
Specifies the size of the tiling image:



target.png
300 × 300 pixels

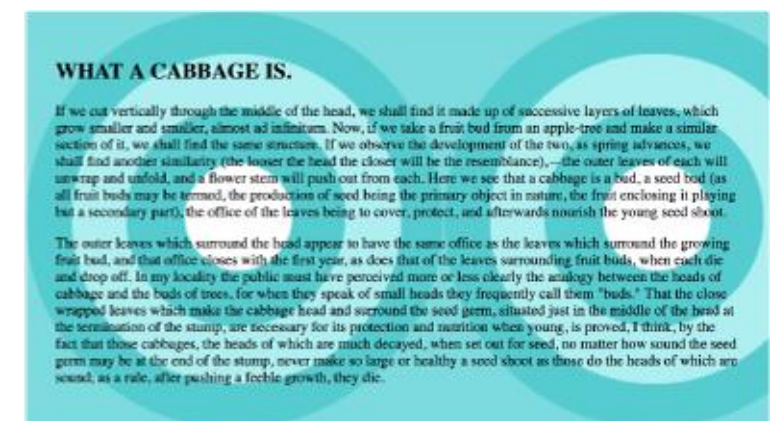


background-size: 600px 300px;



background-size: cover;

The entire background area of the element is covered, and the image maintains its proportions even if it is clipped.



background-size: contain;

The image is sized proportionally so it fits entirely in the element. There may be room left over for tiling (as shown).

Shorthand background Property

background

Values:

background-color background-image background-repeat background-attachment background-position background-size

- Specifies all background properties in one declaration

```
background: white url(star.png) no-repeat top  
center fixed;
```

- Properties are optional and may appear in any order
- Properties not represented reset to their defaults—**be careful it doesn't overwrite previous background settings.**

Multiple Background Images

You can place more than one background image in a single image (separated by commas):

```
body {  
  background:  
    url(image1.png) left top no-repeat,  
    url(image2.png) center center no-repeat,  
    url(image3.png) right bottom no-repeat;  
}
```

Gradient Fills

- A **gradient** is a transition from one color to another.
- **Linear gradients** change colors along a line.
- **Radial gradients** start at a point and spread outward in a circular or elliptical shape.
- You can generate a gradient image for use as a background using **linear-gradient()** and **radial-gradient()** notation.

Example:

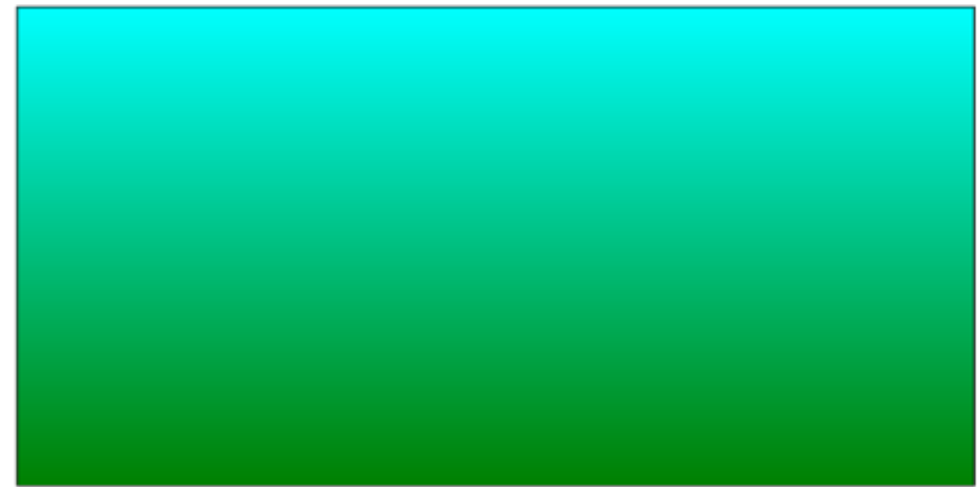
```
#banner {  
  background-image: linear-gradient(180deg, aqua, green);  
}
```

Linear Gradient

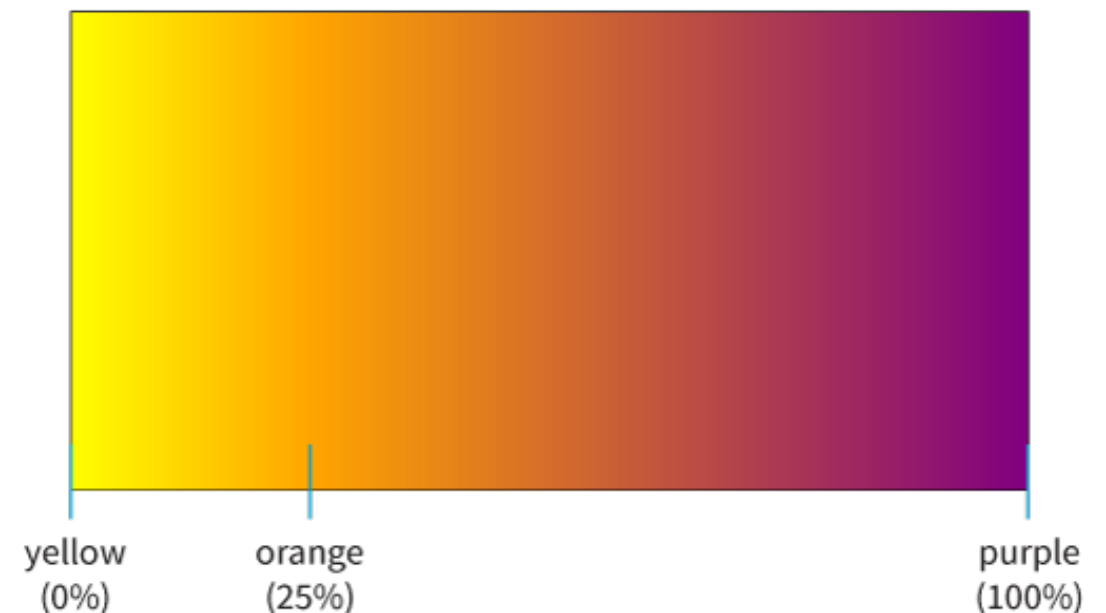
The `linear-gradient()` notation provides the angle of a gradient line and the colors the line passes through.

It is specified in degrees (deg) or keywords (to top, to right, to bottom, to left).

```
linear-gradient(180deg, aqua, green);  
or  
linear-gradient(to bottom, aqua, green);
```



```
linear-gradient(90deg, yellow, orange 25%, purple);
```



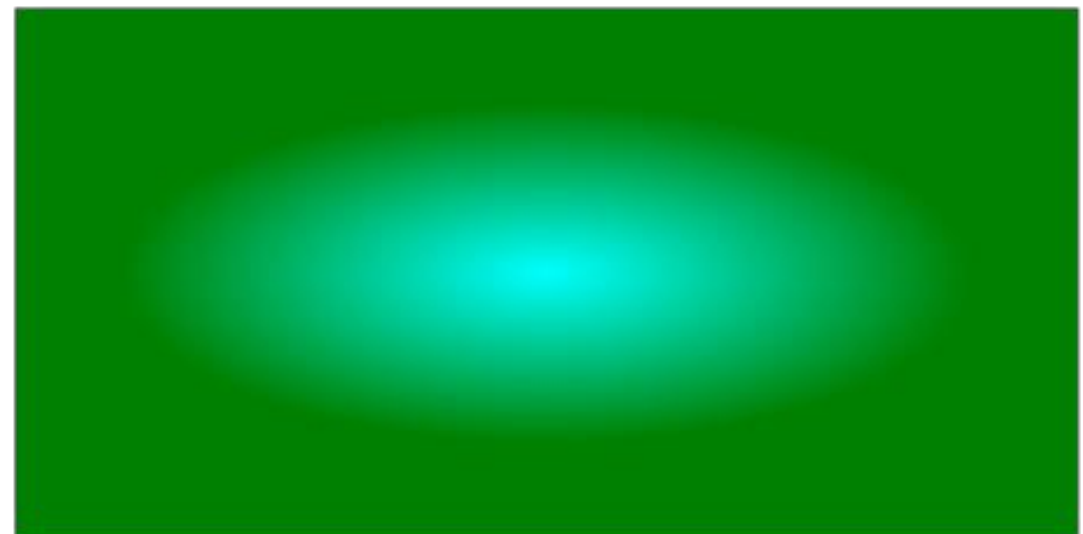
Radial Gradients

The `radial-gradient()` notation provides the color values and optional size, shape, and position information.

```
radial-gradient(yellow, green);
```



```
radial-gradient(200px 80px, aqua, green);
```



More Selector Types

Pseudo-class selectors

Pseudo-element selectors

Attribute selectors

Pseudo-Class Selectors

Treat elements in a certain state as belonging to the same class

Link Pseudo-classes

:link Applies style to unvisited (unclicked) links

:visited Applies style to visited links

User Action Pseudo-classes

:focus Applies when element is selected for input

:hover Applies when the mouse pointer is over the element

:active Applies when the element (such as a link or button) is in the process of being clicked or tapped

Pseudo-classes (cont'd)

Pseudo-classes must appear in the following order:

```
a { text-decoration: none; } /* turns underlines off for all links */  
a:link { color: maroon; }  
a:visited { color: gray; }  
a:focus { color: maroon; background-color: #ffd9d9; }  
a:hover { color: maroon; background-color: #ffd9d9; }  
a:active { color: red; background-color: #ffd9d9; }
```

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

a:link

Links are maroon and not underlined.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

a:focus

a:hover

While the mouse is over the link or when the link has focus, the pink background color appears.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

a:active

As the mouse button is being pressed, the link turns bright red.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

a:visited

After that link has been visited, the link is gray.

More Pseudo-Class Selectors

Structural pseudo-classes

These allow selection based on where the element is in the structure of the document (the document tree):

- `:root`
- `:empty`
- `:first-child`
- `:last-child`
- `:only-child`
- `:first-of-type`
- `:last-of-type`
- `:only-of-type`
- `:nth-child()`
- `:nth-last-child()`
- `:nth-of-type()`
- `:nth-last-of-type()`

OVERVIEW

Input pseudo-classes

These selectors apply to states that are typical for form inputs:

- `:enabled`
- `:disabled`
- `:checked`

Location pseudo-classes (in addition to `:link` and `:visited`)

- `:target` (fragment identifier)

Linguistic pseudo-class

- `:lang()`

Logical pseudo-class

- `:not()`

```
<!DOCTYPE html>
<html>
<head>
<style>
p:first-child {
  color: blue;
}
</style>
</head>
<body>
```

```
<p>This is some text.</p>
<p>This is some text.</p>
```

```
</body>
</html>
```

This is some text.

This is some text.

first-child: Select
and style every
<p> element that
is the first child of
its parent

This *heading*

I am a *strong* person. I am a *strong* person.

I am a *strong* person. I am a *strong* person.

```
<!DOCTYPE html>
<html>
<head>
<style>
p i:first-child {
  color: blue;
}
</style>
</head>
<body>
<h1>This <i>heading</i></h1>
<p>I am a <i>strong</i> person. I am a <i>strong</i>
person.</p>
<p>I am a <i>strong</i> person. I am a <i>strong</i>
person.</p>

</body>
</html>
```



```
<!DOCTYPE html>
<html>
<head>
<style>
p:last-child {
  background: red;
}
</style>
</head>
<body>
```

The first paragraph.

The second paragraph.

The third paragraph.

The fourth paragraph.

```
<p>The first paragraph.</p>
<p>The second paragraph.</p>
<p>The third paragraph.</p>
<p>The fourth paragraph.</p>

</body>
</html>
```



```
<!DOCTYPE html>
<html>
<head>
<style>
p:nth-child(2) {
    background: red;
}
</style>
</head>
<body>
```

The first paragraph.

The second paragraph.

The third paragraph.

The fourth paragraph.

```
<p>The first paragraph.</p>
<p>The second paragraph.</p>
<p>The third paragraph.</p>
<p>The fourth paragraph.</p>

</body>
</html>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
p:lang(it) {
```

```
  background: yellow;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<p>I live in Italy.</p>
```

```
<p lang="it">Ciao bella!</p>
```

```
</body>
```

```
</html>
```

I live in Italy.

Ciao bella!

Pseudo-Element Selectors

Applies styles to elements not explicitly marked up in the source.

::first-line

Applies a style to the first line of an element:

```
p: first-line {letter-spacing: 9px;}
```

::first-letter

Applies a style to the first letter of an element:

```
p: first-letter { font-size 300%; color: orange; }
```

WWF's Mission Statement

```
<!DOCTYPE html>
<html>
<head>
<style>
p::first-line {
  background-color: yellow;
}
</style>
</head>
<body>
```

```
<h1>WWF's Mission Statement</h1>|
<p>To stop the degradation of the planet's natural
environment and to build a future in which humans live
in harmony with nature, by; conserving the world's
biological diversity, ensuring that the use of renewable
natural resources is sustainable, and promoting the
reduction of pollution and wasteful consumption.</p>
```

To stop the degradation of the planet's natural environment and to build a future in which humans live in harmony with nature, by; conserving the world's biological diversity, ensuring that the use of renewable natural resources is sustainable, and promoting the reduction of pollution and wasteful consumption.